Lec 1: Product Analytics

Product strategy &

Business Acumen

– Intro session –

= 9:03 PM =

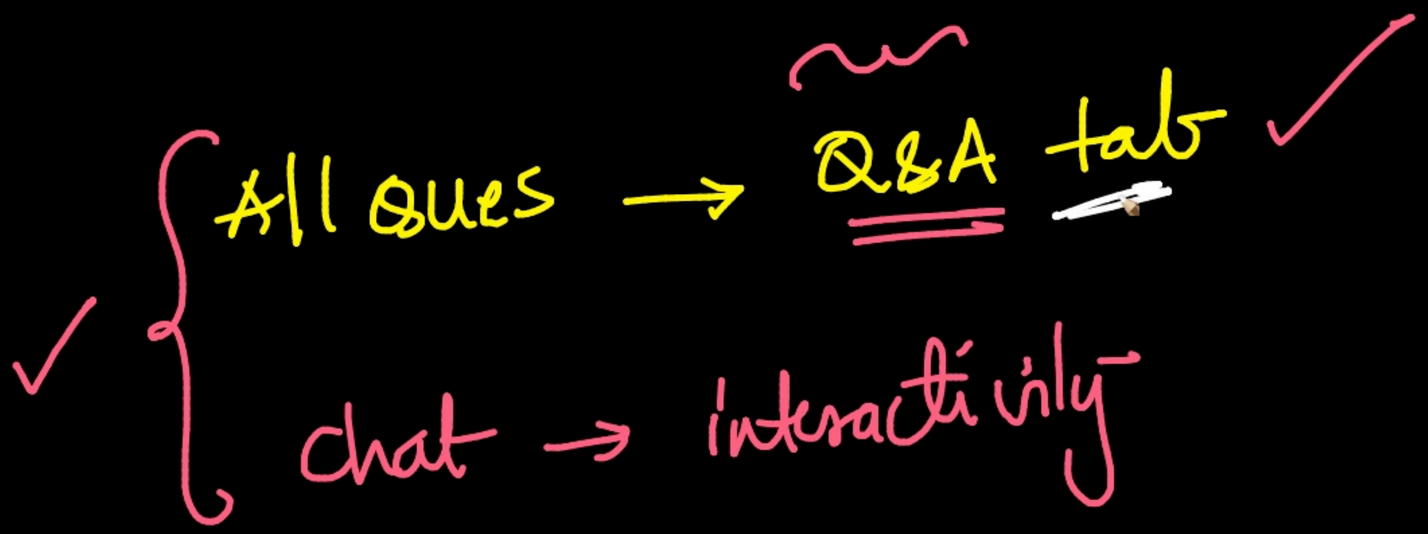
– Application oriented

– Interactivity & Participation

– Business thinking ....

– Asking the 'right' questions

Frameworks


All ques → Q&A tab
chat → interactivity

Rest of the module:

- Product metrics & design → Social media, Fitness, Banking
- Root Cause Analysis → Ecommerce (Myntra), Cabs (Uber)
- ✓ RFM models
- ✓ Customer Segmentation
- ✓ Guestimates → online-gaming
- A/B Testing
  - Mse: Airbnb; Airline bookings ...

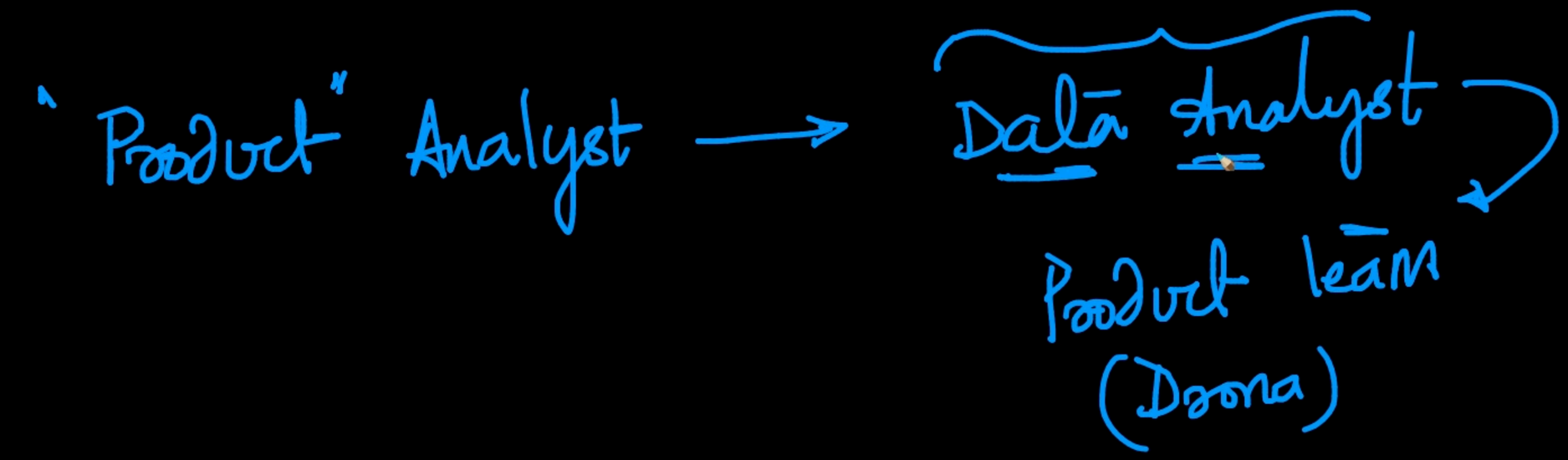
DAU → Prob & Applied stats ✓

↓

hypothesis testing ✓

↳ A/B ...

Microsoft Edge File Edit View History Favourites Tools Profiles Tab Window Help



Microsoft Edge File Edit View History Favourites Tools Profiles Tab Window Help 9:20

obj

1. Analyze a mélúc change
2. Define a mélúc

Microsoft Edge File Edit View History Favourites Tools Profiles Tab Window Help 9:20

Data Analyst →
- extra insights from raw data
- Understand the business

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:34

Product / Business Acumen:

Example ques:

1. Why did Youtube's traffic fall by 5% ?
   - ↳ IPL ? ; outage ? ; more ads ? ; .... (external)

2. 'Save Post' on Face book
   - ↳ design a metrics to measure this feature ?
   - ↳ e.g: Usage of people revisiting the saved posts

Solve it in detail

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:32

3

Product Search on Amazon

↳ what metrics can we use to measure the health of Search

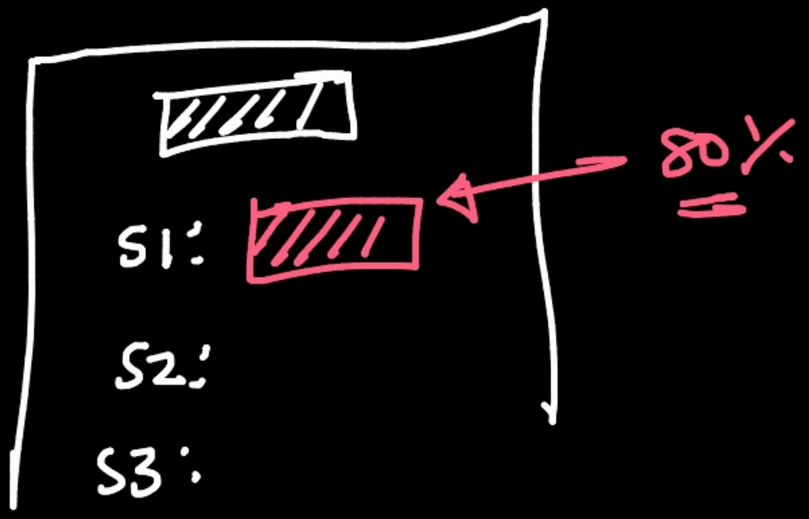
metric

CTR@2: 95%

Algo 2

CTR@1: 80%

Algo 1



Microsoft Edge File Edit View History Favourites Tools Profiles Tab Window Help 9:29

Interview

Judgement: → Structure of your thinking
→ Comprehensively / all-cases
→ Feasability of solution / realistic

Framework: →
1. Clarify ***
2. Plan ...
3. Summarize

10 / 10

4 kinds of problems (non-exhaustive)

4 [1] SOLVE THIS

Product Diagnostics: Analyzing a change in metric

1. Why did Bing Search usage go up by 15% last month?
2. Why did the unique # monthly users reduce on Kukufm ...

MAU : monthly Active users

DAU: daily active users...

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:43

② New feature / product → Measure success metrics

① e.g: "Save Post" → design metrics to measure if this feature is a success

② Lens Kart – AR try-on → ....

solve this!

Chrome File Edit View History Bookmarks Profiles Tab Window Help 9:46

(3) Recommend a new feature launch

↳ should we have Amzn search bar @ the bottom of the page or not? →

→ should the classes be 9:30 to 11:30 PM

→ scan & deliver...

13 / 13

Chrome File Edit View History Bookmarks Profiles Tab Window Help

4 Product Improvement

→ How to improve Google Maps?

→ How to improve Scaler Dashboard?

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:09

1 Product Diagnostics: Analyze a metric change

Case: metric — %age of users who clicked on FB event

Search increased 15% WoW. Investigate this ... — change

Framework: CRIED

- C
- R
- I
- E
- D

I → Internal

D → Data

E → External

C → Clarify

R → rule out

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:09

(a) Clarify

↳ Facebook event Searches ?

(or) FB Search bar

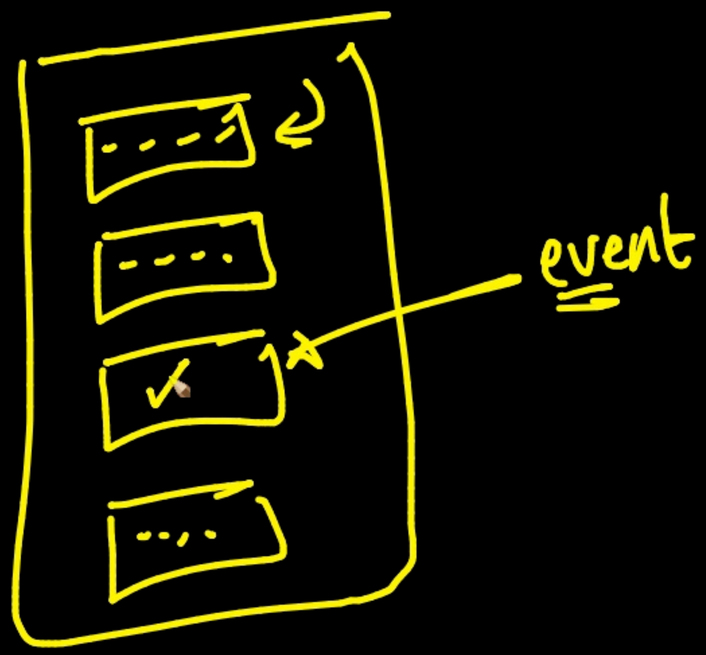


Events



→ (15%) increase WoW

** Metric = # users who click on an event - result / # users who searched


event

15% 1000 increase

↓

Compared to → last week?

→ avg of last 4 weeks?

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:08

CLARITY → eqn. of the metric
↑ or ↓ (over what?)
→ everything...

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:21

✓ Rule-out: technical glitches / outliers / bugs...

1. Counting some users / clicks twice ....
2. Outliers: Spring-break / Summer-vacation / Christmas / ... Diwali
3. Data pipeline / SQL bug: ...

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:15

Internal Data:

TROPICS

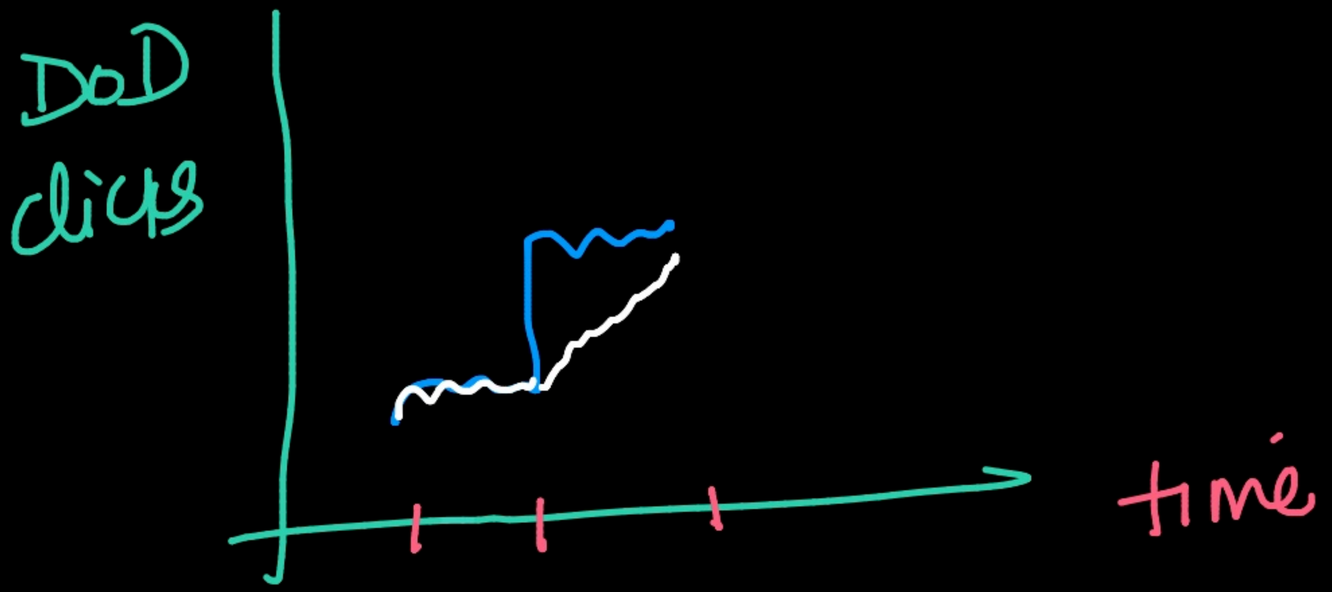
- Time
- Regim
- Other related features effected

→ Platform

Cannibilization

Segmentation

Time → 15% WoW → sudden or gradual

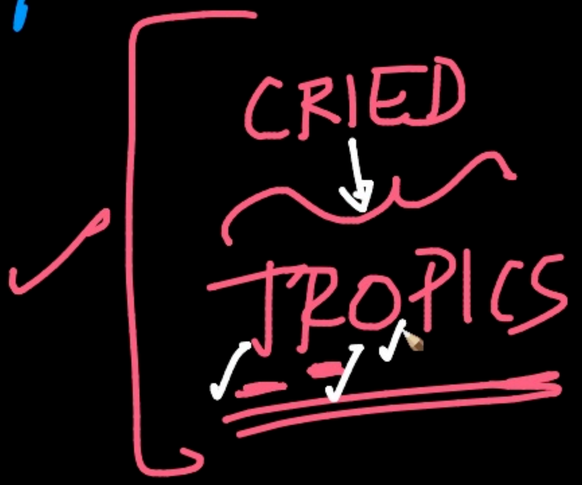


slice on time

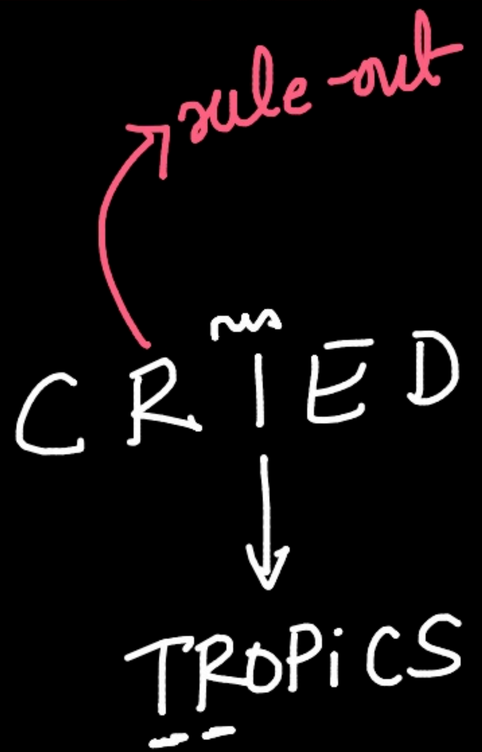
Region: US / India / ... BLR / Hyd / SF ...

or global

→ slicing by geography



Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:22



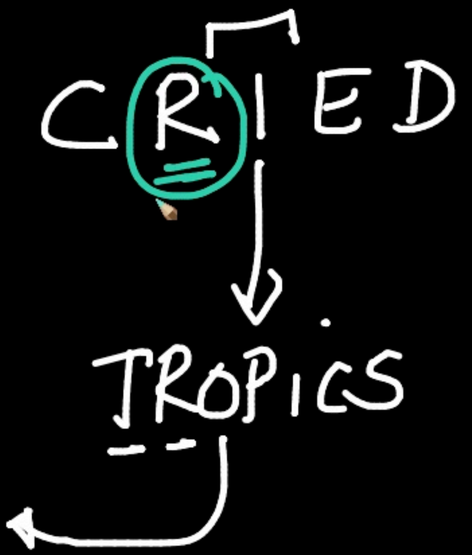
Chrome File Edit View History Bookmarks Profiles Tab Window Help

Q: other related features effected

↳ Posting more about these events : FB / Instagram ...

#ARR concert

↳ more ads for these events



25 / 27

P: platform

→ ios / Android / web → Mac, windows

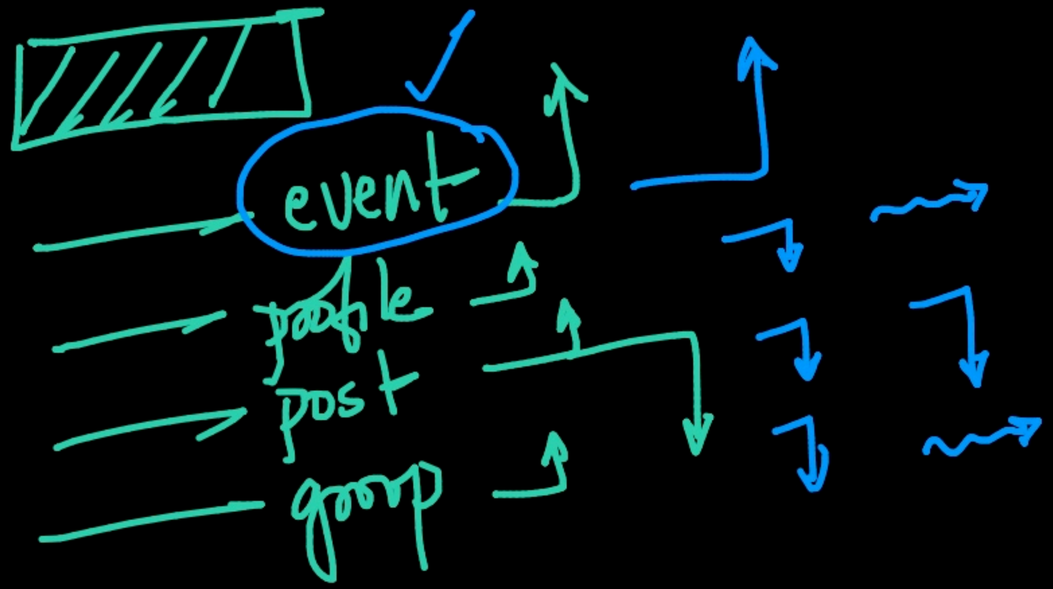
(slice by platform)

C: Cannibalization

Wow clicks on events ↑ 15%

1 Million users ↓ 5 million clicks

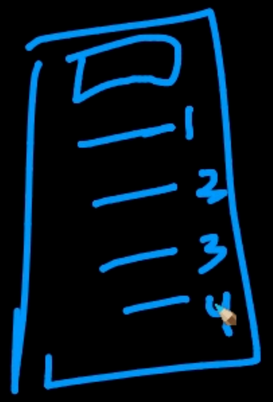
Is there a ↓ somewhere else?

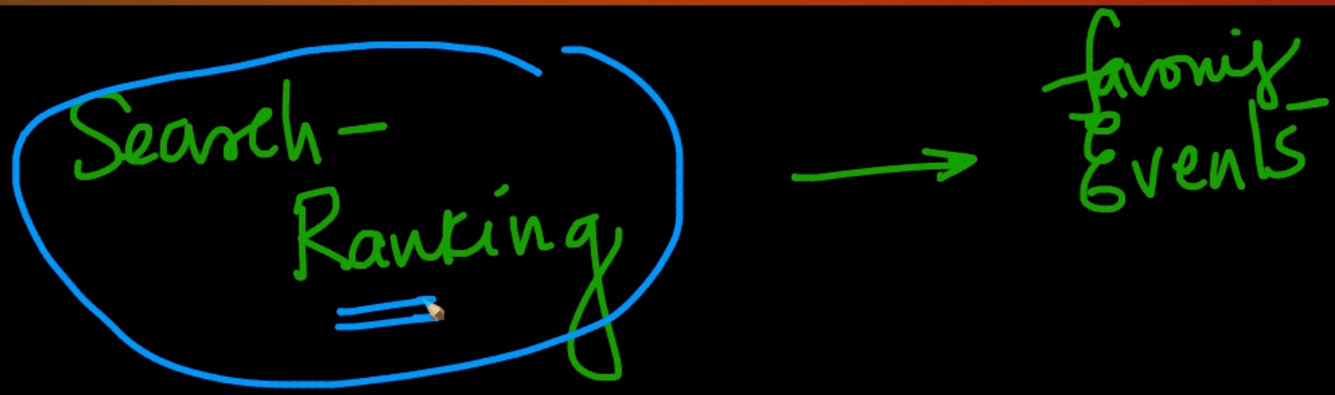


profiles ↓

groups ↓

posts




Search-
Ranking
favorite
Events

Ethnicity → ....

Age →

Gender →

Casual or Addicted (Regular/) users

New users or Existing users

User - Interests → Travel ..
DS & ML ..
Music ...

TROPICS

Segmentation

Language → ..

Education → ...

Job → ...

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:39

CRIED

↓

TROPICS

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:42

External - Data

↳ more tweets about Events (trending hashtags)

↳ Press Release : → $1 off ...

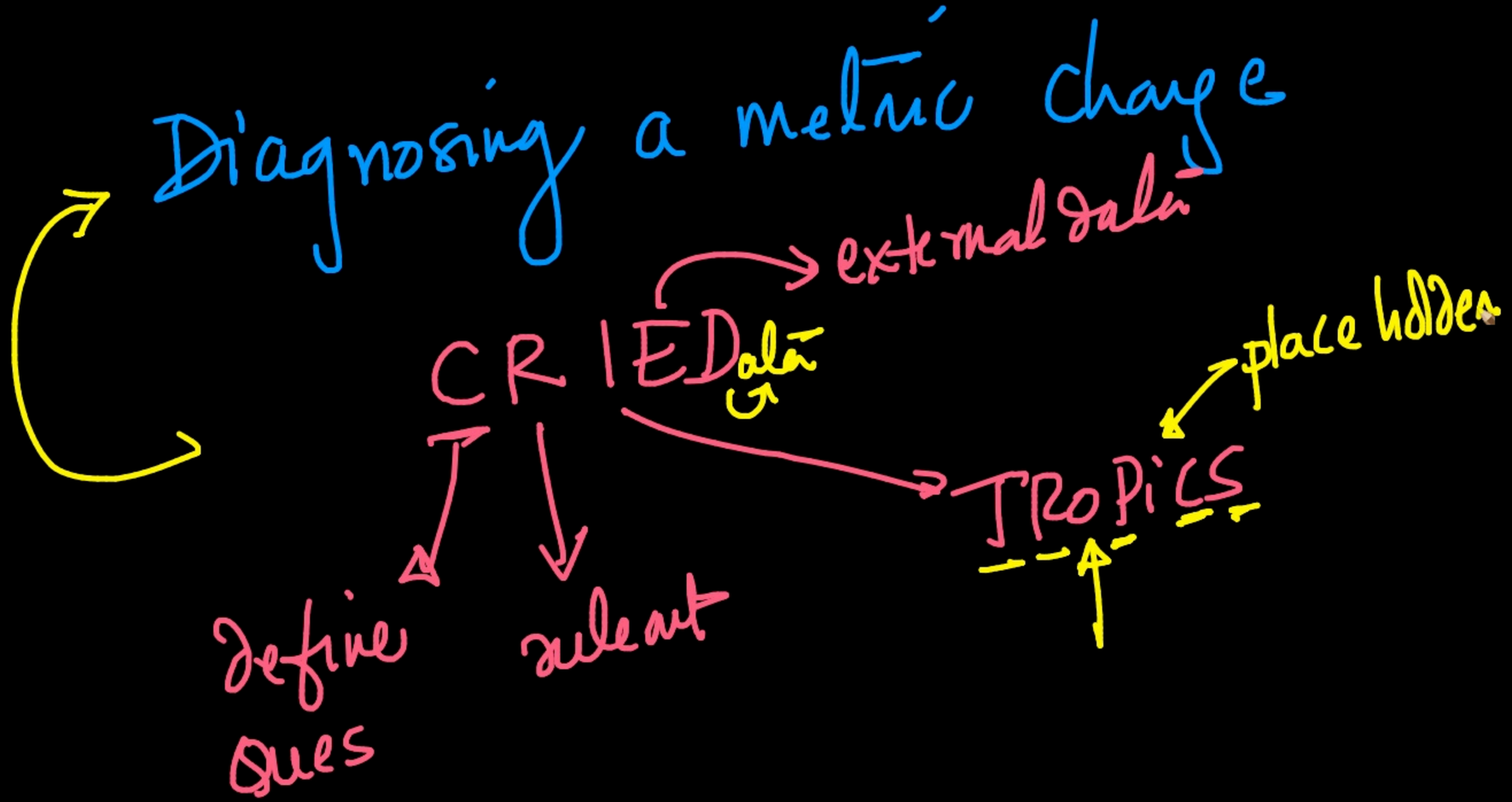
→ Policy changes → ...

→ ...

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:43

Recap:

Diagnosing a metric change



Chrome File Edit View History Bookmarks Profiles Tab Window Help 11:09

2

Define metrics to measure feature / product

↓ "Save for later" feature on Facebook

↓

1. Clarify ← Ques & who is this for?
2. Business Goal ↔ ... .
3. Metrics
4. Summarize

next class

Chrome File Edit View History Bookmarks Profiles Tab Window Help 10:59

Clarify:

↳ what does this feature apply to?

↳ Posts / Images / Links / Pages / Groups / Events ✓
messages / ads / ...

↳ how many can be saved? ✓

↳ expiry of saved posts?

→ accessibility ...

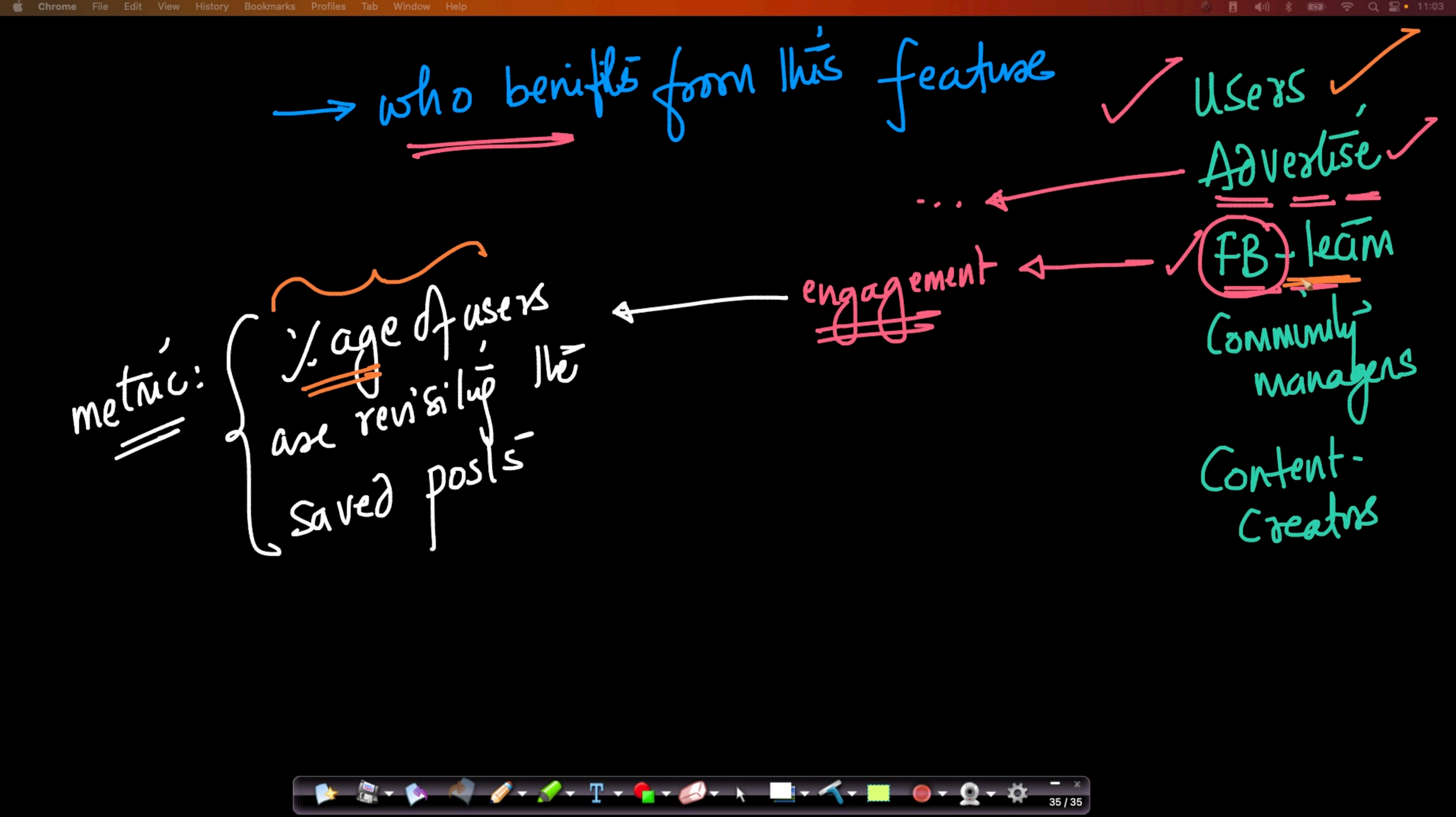
→ grouped?

→ reminded or not?

~~x~~ offline or not?

→ notifications ...

⋮


→ who benifits from this feature
Users
Advertise
...
FB team
Community managers
Content-Creators
engagement
metric:
% age of users are revisiting the saved posts

Q

Diagnose

CRIED

TROPICS

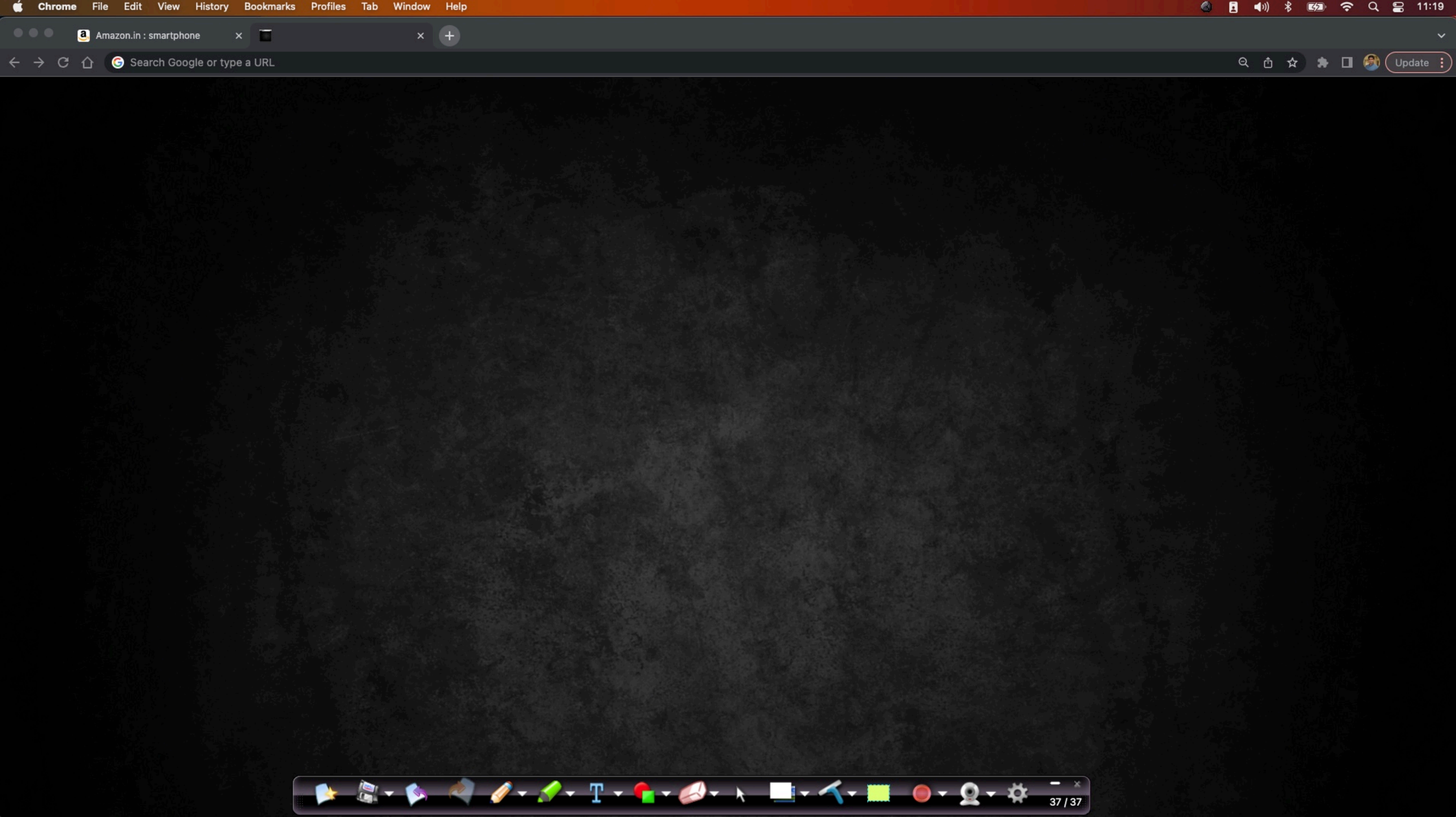
- Structured
- exhaustively
- realistic / feasible

y, inc

R_1
I_2
:
"
;

χ^2 test..

Time Series plot


Chrome File Edit View History Bookmarks Profiles Tab Window Help
11:19
Amazon.in : smartphone
Search Google or type a URL
Update
37 / 37